

Porter Cook

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GitHub Portfolio: <https://porterrockwell.github.io/>

CURRICULUM VITAE

PROFESSIONAL SUMMARY

Mechatronic engineer and Visiting Instructor of Practice in Industrial Robotics at Utah Tech University with hands-on experience in PLC controls, industrial robot programming (FANUC, Yaskawa, Universal Robots), and automated test systems for aerospace manufacturing. Helped develop and restructure the Industrial Robotics curriculum, introduced FANUC ROBOGUIDE offline programming to the program, and helped build a PLC-to-robot trainer for teaching I/O, digital logic, and Ethernet communication. Responsible for the electrical and controls design of an industry-partnered senior capstone project with Lovesac.

EDUCATION

Utah Tech University - St. George, UT | Bachelors of Science

Mechatronic Engineering

Coursework: Industrial Networking, SCADA, Industrial Wiring, Robotics, Programmable Logic Controllers, Computer Aided Manufacturing, Computer Aided Design, Ladder Logic, Systems Design, Additive Manufacturing

Utah Tech University - St. George, UT | Associates

Pre-Engineering, May 2023

SKILLS

PLC Programming: Ladder Logic & Structured Text	FANUC, Yaskawa & Universal Robots
Beckhoff PLC & TwinCAT 3	FANUC ROBOGUIDE & HandlingTool
Productivity Suite & SCADA	C#, C++, SQL, Python & HTML
Group I/O & Ethernet Communication	WPF Applications & SQL Databases
Industrial Wiring & Troubleshooting	Fusion 360, SolidWorks, AutoCAD Electrical, Onshape
Digital Logic	CNC, CAD/CAM & 3D Printing
Teaching & Lab Safety	Problem-Solving & Teamwork

WORK HISTORY

Visiting Instructor of Practice

April 2026 – Present

Utah Tech University, Industrial Robotics

- Helped develop a new Industrial Robotics curriculum and restructured the course (practical exposure, then theory, then application on physical equipment) to teach static and inertial forces, I/O, and Ethernet communication alongside robot programming and operation
- First instructor to introduce FANUC ROBOGUIDE into the curriculum, giving students offline programming and virtual workcell experience
- Helped develop a PLC trainer pairing an AutomationDirect P1000 PLC with a FANUC LR Mate 200iD robot through CRMA58 terminal blocks to teach group I/O, digital logic, and Ethernet communication between a PLC and a robot controller
- Teaching MTRN 2200 and MTRN 2205 for the first time, with emphasis on safe, practical industrial wiring and troubleshooting
- Implemented lab safety protocols for student use of industrial robotics equipment
- Lead instructor for hands-on lab instruction, guiding students through robot programming and operation

- Pursuing FANUC CERT Instructor Certification through FANUC America to become a FANUC-certified robotics instructor

Automation Technician

March 2026 – April 2026

IME Automation

- Gained hands-on skills in end-of-arm tooling and robotic material handling
- Contributed to a single automated system project from development through completion, strengthening skills in automation controls, integration, and troubleshooting

Automation Engineering Intern

Aug 2024 – Sept 2025

Baxter Aerospace

- Developed the HVX (High Voltage Automated Test) system to test aerospace wire harnesses for continuity and perform high voltage testing
- Created automated processes, including wire testing equipment for Quality Assurance
- Integrated PLC control for sensor data collection and process automation
- Programmed Yaskawa, Universal Robots, and FANUC robots to automate manufacturing processes
- Developed C# WPF applications linked to SQL databases for MES work instructions and assembly lines
- Worked in Research and Development on several beta automation projects

Maker Space Intern

August 2022 – May 2023

Innovation Plaza, Utah Tech University

- Trained individuals to use laser engravers, CNCs, and 3D printers at the Makerspace
- Fixed and installed safety features on machinery such as E-stops and circuit breakers
- Served as lead operator for the Stinger CNC machine, modified g-code, and created tool paths in Fusion 360
- Worked with individuals to design parts in Fusion 360 and SolidWorks, including prosthetics and business-related items

Engineering Intern

2022

Utah Tech University, Engineering Department

- Worked with Professor Dave Christensen on engineering displays and design projects
- Designed 3D models in Fusion 360, SolidWorks, and Visio; organized inventory and supported SLA printer maintenance

EDUCATIONAL AND ACADEMIC PROJECTS

Lovesac Material Dispensing System - Senior Capstone

- Industry partnership with Lovesac through Utah Tech University; responsible for the electrical and controls side end-to-end
- Designed the HMI, wrote the ladder logic, and designed and wired the main enclosure (AC and 24VDC distribution, PLC I/O, dual VFD drive bays with dynamic braking resistors)
- Trained polynomial regression and neural-network surrogate models (RMSE 0.346 and 0.303 lbs) to find repeatable frequency and time settings; all four optimized settings landed within 0.05 lbs of target

PLC / Robot Ethernet Trainer - AutomationDirect P1000 and FANUC LR Mate 200iD

- Pairs a PLC with an industrial robot through CRMA58 terminal blocks to teach group I/O, digital logic, and Ethernet communication; used in the Industrial Robotics course

PLC Training Cabinet - Programmable Logic Controllers Final

- Constructed a PLC training cabinet from blueprints as lead mechatronic engineer: P3000 PLCs, safety relays, HMI screens, Ethernet junction boxes, motion controllers, encoders, and NPN/PNP sensors

AR4 6-Axis Robotic Arm - Systems Design Final

- Designed the assembly in Fusion 360 with a Teensy 4.1 controller, standalone and PLC-triggered control modes, and a control cabinet for AMC motor control integration

Additional Projects

- Fire Knife (Additive Manufacturing Final): SolidWorks assembly with bill of materials, CNC-machined retainer, custom valve body
- Personal PLC Cabinet: Direct Logic 05 fixed and Direct Logic 205 modular PLCs for prototyping
- BLDC Motor for regatta competition: final-course team project with Baylee Schumacher

RESEARCH AND GRANT WORK

Adaptive Manufacturing and Automation Testbed (AMAT) - helped prepare a grant proposal at Utah Tech for a statewide, automation-ready machining testbed that develops, validates, and transfers production-ready processes to Utah manufacturers.

CERTIFICATIONS AND TRAINING

- FANUC Academy - HandlingTool Operations & Programming (4 Days, Virtual), May 21, 2026
- Beckhoff E-Learning - TwinCAT 3: Getting Started, May 15, 2024
- New Faculty Seminar - Certificate of Participation
- FANUC CERT Instructor Certification - in progress, FANUC America

LEADERSHIP AND SERVICE

FSY Counselor

May 2022 – August 2022

Church of Jesus Christ of Latter-day Saints

- In charge of the welfare and safety of FSY attendees; coordinated lesson plans, housing, food, and event safety with a team of counselors

Missionary

May 2019 – May 2021

Church of Jesus Christ of Latter-day Saints

- Taught and led a group of 12 missionaries in Las Vegas, NV; also served in Accra, Ghana
- Monitored team performance, recommended improvement plans, and served as social media specialist for a local group

REFERENCES

Available in the accompanying References document and on request.